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Thermochronometry and palaeomagnetism of the Archaean Nelshoogte Pluton, South Africa

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SUMMARY

$^{207}Pb/^{206}Pb$ single-grain zircon, $^{40}Ar/^{39}Ar$ single-grain hornblende and biotite, and $^{40}Ar/^{39}Ar$ bulk-sample muscovite and biotite ages from the Nelshoogte trondhjemite pluton located in eastern Transvaal, South Africa, show that this granitoid had a protracted thermal history spanning 3213 ± 4 Ma to about 3000 Ma. Whole-rock $^{40}Ar/^{39}Ar$ ages from cross-cutting dolerite dykes indicate that these were intruded at about 1900 Ma. There is no evidence of this or other, later events significantly affecting the argon systematics of the minerals from the pluton dated by the $^{40}Ar/^{39}Ar$ method.

The pluton has a well-defined palaeomagnetic pole which is dated at 3179 ± 18 (2σ) Ma by $^{40}Ar/^{39}Ar$ dating of hornblende. This pole (18°N, 310°E, $A_{95} = 9°$) yields a palaeolatitude of 0°, significantly different from other Archaean poles from the Kaapvaal Craton. The palaeolatitude difference implies that there was significant apparent polar wander during the Archaean. A second, overprinting magnetization seen in the pluton is also seen in the lower-Proterozoic dolerite dykes, and is consistent with other lower-Proterozoic (2150–1950 Ma) poles for southern Africa.

Key words: Archaean, geochronology, palaeomagnetism, South Africa, thermochronology.

INTRODUCTION

For many years, the interpretation of Archaean palaeomagnetic data has been hampered by the lack of precise ages to assign to particular palaeomagnetic poles. Refinements in U–Pb and $^{40}Ar/^{39}Ar$ dating methods make it possible to assign ages to palaeomagnetic poles better. With more precisely dated poles, questions regarding Archaean apparent polar wander and magnetic field evolution (e.g. dipole versus non-dipole effects and palaeointensity) can be addressed more fully.

Evans (1967) first suggested that southern Africa experienced significant apparent polar wander in the late Archaean and early Proterozoic, on the basis of poles from the Modipe Gabbro (Evans & McElhinny 1966), the Great Dyke (McElhinny & Gough 1963) and the Gaborone Granite (Evans 1967). Hale (1987) used these data and a pole from the Komati Formation in the Barberton Greenstone Belt (Hale & Dunlop 1984) to support a hypothesis of no apparent polar wander in the Archaean. This conclusion was based on the observation that the Komati Formation pole and the Modipe Gabbro pole indicate polar palaeolatitudes for the Kaapvaal Craton in both the early (Komati) and late (Modipe) Archaean. Subsequent studies of late-Archaean units such as the 2870 Ma Usushwana Complex (Layer *et al.* 1988a) and the 2687 Ma Mbabane Granite Pluton (Layer *et al.* 1989) of Swaziland provide evidence of intermediate and equatorial latitudes, respectively. In addition, Evans, Beukes & Kirschvink (1997) report evidence of low latitudes for the Palaeoproterozoic (2222 Ma) Ongeluk lavas of the Transvaal Group. These data indicate that there has been significantly more apparent polar wander in the Archaean and early Palaeoproterozoic than was apparent with the limited data available to Hale (1987).

Here we report the results of a palaeomagnetic and geochronologic study of the trondhjemitic Nelshoogte Pluton located in eastern Transvaal, South Africa. The $^{207}Pb/^{206}Pb$ zircon age of this pluton is about 3213 Ma, clearly Archaean, and a stable pole obtained from it provides further evidence for significant Archaean apparent polar wander, contrary to Hale's (1987) hypothesis.

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GEOLOGICAL OVERVIEW

The Nelshoogte Pluton (NHP) is one of several plutons of tonalitic–trondhjemitic composition which surround the Barberton Greenstone Belt (BGB) in the eastern Kaapvaal Craton, South Africa (Fig. 1). It is exposed as a semicircular body some 20 km in diameter with the (presumed) western half of the pluton covered by the Transvaal Supergroup. In the eastern Transvaal, the Transvaal Supergroup and related rocks form a thick sequence with an age range from 2552 ± 11 Ma at the base to 2432 ± 31 Ma near the top of the sequence (Barton *et al.* 1994). The NHP is bordered on the east by the BGB, on the north by the Kaap Valley Pluton (KVP) [zircon age 3226 ± 4 (2σ) Ma (Tegtmeyer & Kröner 1987; Kamo, Davis & de Wit 1990; Layer, Kröner & York 1992)] and on the south by the Kees Zyn Doorns syenite and the Badplaas trondhjemitic cell (Fig. 1). The contact with the KVP is not well exposed. It is presumed, on the basis of isotopic evidence and regional geological models, that the NHP intrudes the KVP (Robb *et al.* 1986). The contact between the NHP and the schists of the BGB to the east is a sheared margin.

The NHP is generally a coarse-grained trondhjemite composed of plagioclase, quartz, microcline, biotite, epidote, muscovite and minor opaques. The biotite crystals are subhedral to euhedral, and define the foliation. They are pleochroic, light greenish yellow to reddish brown. The biotites are variably altered to green chlorite and epidote. The average alteration to chlorite may be 50 per cent or more. Muscovite occurs as coarse prisms up to 1–2 mm in length, invariably associated with the biotite, and as fine sericite needles in the saussuritized plagioclase. Quartz-rich and feldspar-rich bands define a concentric foliation. This foliation is accentuated by patchy chloritized biotite that parallels this crude mineralogical layering.

The NHP is cut by a northwest–southeast-trending joint pattern. Intruded along this subvertical pattern are doleritic dykes, presumed to be early Proterozoic in age on the basis of cross-cutting relationships (Robb *et al.* 1986; Hunter & Reid 1987). In the course of field sampling, a second set of microgabbro dykes (MGDs) with a trend similar to the dolerite dykes (henceforth referred to as DDs) but with a coarser grain size was identified. The MGDs show evidence of slow cooling, and there are signs of reaction between the pluton and the dyke along the margin.

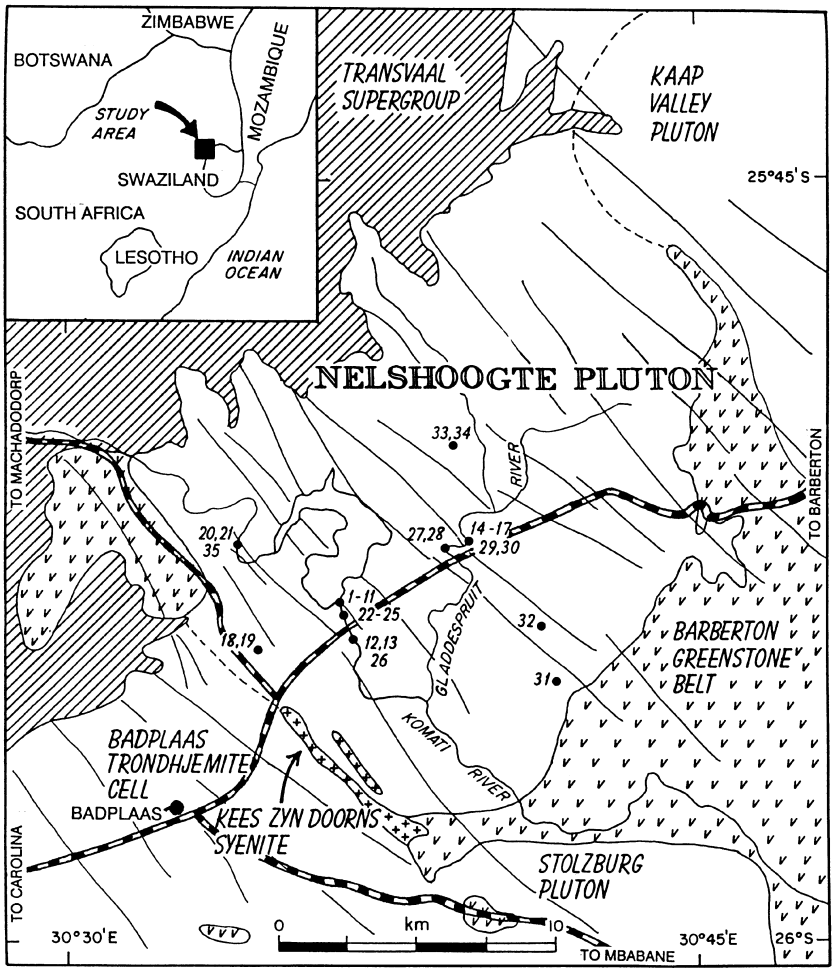


Figure 1. Generalized geological map of the Nelshoogte Pluton showing palaeomagnetic sampling sites (circles) and northwest–southeast-trending cross-cutting dykes (thin lines). Tonalite–trondhjemite plutons (Kaap Valley, Nelshoogte, Badplaas and Stolzberg) are shown without shading, and the inferred boundaries between them are shown by dashed lines. The 'v's denote rocks of the Barberton Greenstone Belt, and the '+'s the Kees Zyn Doorns syenite.

The overlying Transvaal Supergroup sediments dip gently ($<5°$) to the west, implying that the NHP has not been tilted significantly since about 2500 Ma. The 3104 ± 4 Ma Mpuluzi Batholith (Kamo *et al.* 1990) located 20 km south of the NHP is a large sheet-like batholith and also shows no evidence of internal deformation or tilting (Anhaeusser & Robb 1983). Thus, it is likely that the NHP has not experienced major tilting since 3100 Ma, and in the following discussion we assume no post-magnetization tilt. However, we cannot conclusively rule out that tilt may have occurred prior to the deposition of the Transvaal Supergroup.

AGE DETERMINATIONS

A Rb–Sr whole-rock isochron age of 3180 ± 75 Ma and a Rb–Sr biotite whole-rock age of 2993 ± 60 Ma have been determined for the NHP (Barton *et al.* 1983). A model $^{207}Pb/^{206}Pb$ age of 3220 ± 80 Ma was also reported for apatite separates from the pluton (Barton *et al.* 1983).

Pb–Pb single-grain dating

Zircon grains from a sample labelled 'AGD', from sampling site 3 (Fig. 1), were dated by the single-grain evaporation technique (Kober 1986, 1987). In this method, chemically untreated zircons about 100–200 microns in size are embedded in the evaporation filament of a rhenium double-filament arrangement, and Pb is then evaporated in a mass spectrometer during the heating of the grain. Details of the technique are reported by Kober (1986, 1987) and Kröner & Todt (1988). Kober (1986) has shown that the Pb components with the highest activation energy normally reside in the undamaged crystalline zircon phase, which shows no post-crystalline Pb loss and therefore yields concordant $^{207}Pb/^{206}Pb$ ages. The evaporation technique has been tested on isotope standards and natural zircons previously dated conventionally and by ion microprobe (see Kröner, Compston & Williams 1989 for details), and the results are identical within error, even for zircons with complex metamorphic histories.

Isotopic measurements were carried out on a Finnigan-MAT 261 mass spectrometer in the Max-Planck-Institut für Chemie in Mainz. Single-grain evaporation of two NHP zircons yielded a combined $^{207}Pb/^{206}Pb$ age of 3213 ± 4 Ma (2σ) (Table 1, Fig. 2), which is consistent with, but more precise than, the previously reported apatite age.

Conventional $^{40}Ar/^{39}Ar$ dating experiments

Two specimens in the trondhjemite were studied by the $^{40}Ar/^{39}Ar$ step heating method. The samples from the Nelshoogte Pluton identified by the numbers BT-1C and BT-1F were petrographically similar and were classified as trondhjemite. In addition, two whole-rock fragments from two cross-cutting doleritic dykes were analysed. All samples were taken from the vicinity of palaeomagnetic sampling site 22 (Fig. 1).

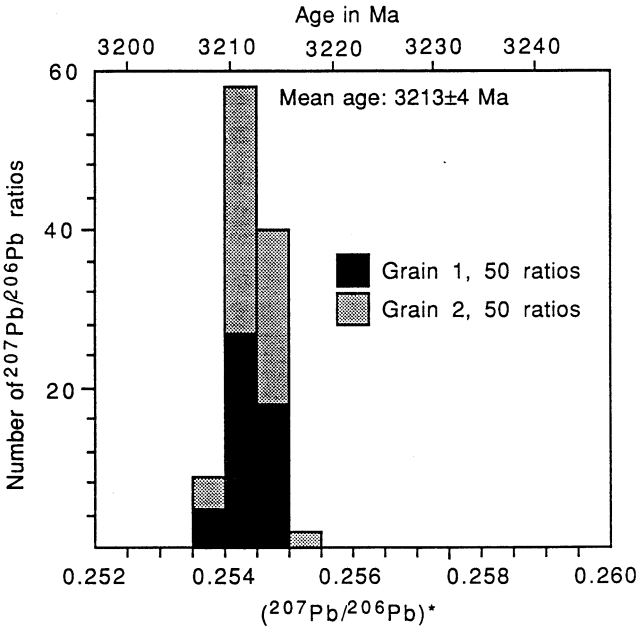


Figure 2. Histogram showing distribution of radiogenic lead isotopes derived from evaporation of zircons from the Nelshoogte Pluton.

Each dolerite rock fragment weighed about 200 mg, and the samples of mineral separates from the trondhjemite weighed approximately 10 mg each. The mineral separates were obtained by crushing the sample to a -70, $+120$ mesh size. Then, by a combination of heavy liquids, a Franz isodynamic separator and hand picking, a pure (99.5 per cent) sample was obtained. The whole-rock fragments and the mineral separates were weighed in small Al foil envelopes. Three samples of the Zartman hornblende standard 'hb3gr' (Zartman 1964) of about 10 mg each were prepared with each set of unknowns. The samples were sent to the reactor in cylindrical Al canisters. The standards were arranged parallel to the length of the cylinder with a 120° spacing.

All cans were irradiated for about 150 h in the enriched-uranium research reactor at McMaster University in Hamilton, Ontario, at position 5C, where variations in the flux gradient are less than 1 per cent (Masliwec 1981). As the standards were all placed symmetrically around the perimeter of the

Table 1. Pb isotopic data from single-grain evaporation of trondhjemite sample AGD.

Grain	Mass scans*	Evaporation temperature (°C)	Mean $^{207}Pb/^{206}Pb$†	$^{207}Pb/^{206}Pb$ age (Ma)
1	50	1495	0.25441 ± 8	3212.4 ± 0.5
2	59	1490	0.25443 ± 8	3212.5 ± 0.5
1&2	109		0.25442 ± 5	3212.5 ± 0.3

* Number of $^{207}Pb/^{206}Pb$ ratios evaluated for age assessment.
† Observed mean ratio, no correction for common Pb since $^{206}Pb/^{204}Pb > 50\,000$. 2σ errors based on uncertainties in counting statistics.

can, a weighted mean of the $^{40}Ar^*/^{39}Ar$ values for the three standards was used in the ensuing calculations.

The mineral separates were analysed by the step-heating method as detailed by Berger & York (1981), using a Lindberg furnace with refined temperature control, to obtain about 15 fractions up to 1200 °C. The majority of the samples were subsequently completely fused by means of a radio-frequency (RF) system capable of reaching 1600 °C. The hb3gr standards were fused in one-step analyses by means of the University of Toronto laser probe (York *et al.* 1981). The whole-rock fragments were step-heated in less detail, with only 10 fractions being collected. These experiments were conducted entirely with the RF system.

Argon isotope ratios in the purified gases of both the samples and the standards were measured with an MS-10 mass spectrometer run in static mode. The mass-spectrometric ratios were then corrected for mass discrimination, sensitivity, neutron interference, radioactive decay of ^{37}Ar and atmospheric contamination, using a program developed by Hall (1982). In all analyses, the decay constants recommended by Steiger & Jäger (1977) (Appendix A) were used to calculate the ages. The analytical precision was determined as described by Hall (1982) and is reported as two standard deviations (2σ). Integrated ages were calculated from a weighted mean of the individual ages calculated for each fraction. The error in the integrated age also includes the error due to the analytical scatter in the standards. Plateau ages were calculated for those regions of the age spectrum where (1) at least three consecutive fractions agreed in age within their 2σ errors, and (2) the cumulative ^{39}Ar release in the segment was at least 30 per cent of the total ^{39}Ar. The age and errors for the plateau were calculated in the same way as the integrated ages.

Muscovite, biotite and microcline separates were obtained from the two Nelshoogte samples and analysed. The results obtained are given in Table 2 and Appendix A. The integrated ages, as well as the plateau ages, calculated for the mineral separates from sample BT-1C are consistently older than those obtained from sample BT-1F.

Age spectra obtained for muscovites from both sites are shown in Fig. 3(a). It can be seen that the age spectrum for sample BT-1F seems more perturbed than that of BT-1C, particularly in the lower-temperature fractions. The plateau ages calculated for these samples are 3107 ± 54 Ma for BT-1C and 3032 ± 8 Ma (2σ) for BT-1F and are statistically distinct. Furthermore, rough estimates of the total K and Ca contents (uncertainties of about 10 per cent) were obtained from the absolute amounts of potassium-derived ^{39}Ar and calcium-derived ^{37}Ar. These estimates yielded 8.4 per cent K and 0.3 per cent Ca for sample BT-1C, while sample BT-1F seems to have suffered some K loss and Ca gain. The K and Ca estimates for this sample were 5.6 per cent and 1.5 per cent, respectively (Fig. 3b). It is possible that the argon systematics have been altered in sample BT-1F, and perhaps this is related to the change in the chemical composition of the mica, or that this difference in the chemical composition generated a less retentive mica.

The biotite age and Ca/K spectra, as can be seen in Figs 3(c) and (d), respectively, follow the same pattern as those for the muscovites. The BT-1C biotite yielded an older, and slightly less perturbed, age spectrum than did the BT-1F biotite. Duplicate analyses performed on BT-1C biotite yielded a plateau age of 3018 ± 17 Ma while a single analysis on the BT-1F biotite yielded a plateau age of 2952 ± 8 Ma. The total K and Ca estimates for these biotites are 6.7 per cent K and 0.3 per cent Ca for sample BT-1C and 3.9 per cent K and 1.2 per cent Ca for sample BT-1F. The difference in ages between the two samples could be due to an increased degree of chloritization in the BT-1F biotite with respect to the biotite from site BT-1C, which is also reflected in the lower K and higher Ca observed in the BT-1F biotite.

The results obtained on the microcline samples yielded much lower integrated ages for both samples. Duplicate analyses for the BT-1C microcline and a single experiment on the BT-1F microcline show staircase spectra, indicative of significant argon loss (Fig. 4). Mathematical modelling of the age spectra following Hall & York (1982) was attempted. However, meaningless and inconsistent results were obtained for both samples. The complicated age spectra indicate that these microclines may have multiple domain sizes and may have experienced a complex thermal history (Lovera *et al.* 1989), and therefore cannot be modelled by simple diffusional loss.

Two whole-rock samples from two different dolerite dykes were analysed (Fig. 5 and Appendix A). Both samples yielded significantly younger ages of about 1900 Ma. Sample KD-15B12 yielded a more perturbed spectrum than did sample KD-4B3. The segment of the age spectrum which was closest to forming a plateau was obtained on sample KD-4B3 for fractions 6 and 7, which yielded ages of 1920 ± 3 and 1953 ± 3 Ma and represent 74 per cent of the total ^{39}Ar released.

Single-grain $^{40}Ar/^{39}Ar$ dating

To complement the conventional $^{40}Ar/^{39}Ar$ study, single hornblende and biotite grains from the MGD at site 21 (Fig. 1) were dated by the single-grain step-heating technique (Layer, Hall & York 1987). Potassium content determinations performed at the University of Alaska Fairbanks found that the hornblende contained 1.180 wt per cent K_2O (0.980 wt per cent K), and the biotite contained 2.647 wt per cent K_2O (2.197 wt per cent K). The low potassium content in the biotite (unaltered biotites generally have approximately 9 wt per cent K_2O), which was also seen in biotites from sample BT-1F, indicates that these biotites may have been severely altered and suffered potassium loss.

The mineral grains and three grains of the Zartman hornblende standard hb3gr were irradiated on aluminium trays, which were loaded into the inlet system for 192 MW h. Each standard was fused in a single step and the *J*-values from the three grains were averaged to determine a *J* for the tray. The mineral grains were step-heated with a continuous laser with a beam size of sufficient diameter to cover the grain completely and ensure uniform heating. The temperature of the grain during heating was monitored using an infrared microscope. Analyses were performed on a VG Micromass 1200S mass spectrometer at the University of Toronto.

Four hornblende grains were step-heated, with one to seven fractions collected per grain (Appendix A). Three of the four grains showed consistent ages and Ca/K values; however, the fourth grain (hbd #3, Table 2) had a slightly younger age and higher Ca/K value (Figs 6a and b), and in the following discussions of the age of the hornblende, this grain is not included. Step-heating of the hornblende revealed that all fractions had the same age within experimental error (Fig. 6a).

Table 2. Summary of $^{40}Ar/^{39}Ar$ results.

Sample	t_i (Ma)	t_p (Ma)	N*	Temp. (°C)†	$\%^{39}Ar$§	$^{37}Ar/^{39}Ar$¶
Bulk samples: Site BT-1C						
Microcline 1	2170±20					
Microcline 2	2317±16					
Biotite 1	2990±17	3018±17	9	785–1065	59	0.0215
Biotite 2	2994±17	3018±17	9	785–1065	60	0.0353
Muscovite	3094±54	3107±54	8	800–1050	80	0.0109
Bulk samples: Site BT-1F						
Microcline	2026±8					
Biotite	2858±8	2952±8	9	800–1050	58	0.152
Muscovite	2997±8	3032±8	7	852–1050	73	0.0986
Cross-cutting dykes (dolerite)						
KD-4B3	1902±6					
KD-15B12	1876±6					
Single grains: microgabbro dyke (MGD)						
Hornblende 1	3185±18		1		100	3.76
Hornblende 2	3179±20	3179±20	7		100	3.88
Hornblende 3	3154±28	3154±28	6		100	4.47
Hornblende 4	3175±28	3175±28	3		100	3.85
1+2+4		3179±18				3.84
Biotite 1	2857±22	2891±21	7	700–1100	80	0.0183
Biotite 2	2822±20	2904±22	9	700–1600	66	0.0301

t_i: integrated age; t_p: plateau age. All errors reported are 2σ.
* Number of fractions involved in the plateau calculation.
† Temperature range over the plateau fractions.
§ Percentage of ^{39}Ar released on the plateau.
¶ Ratio of ^{37}Ar to ^{39}Ar for the plateau fractions.

There was no evidence of argon loss or excess argon in the low-temperature release fractions as has been seen in other single hornblende grains (e.g. Layer *et al.* 1989). In calculating the plateau ages for these grains, all fractions could be used, hence the plateau ages equal the integrated ages. The three hornblende grains have a combined $^{40}Ar/^{39}Ar$ plateau age of 3179±18 Ma (Table 2).

Two biotite grains from this site have age spectra reminiscent of the bulk-sample spectrum from site BT-1F in the pluton (Figs 6c and d and Appendix A). These spectra are disturbed and yield poorly constrained plateau ages of 2891±21 Ma and 2904±22 Ma, respectively, similar to, but slightly younger than, the plateau from the BT-1F biotite.

INTERPRETATION OF THE GEOCHRONOLOGY

We interpret the new ages from the Nelshoogte Pluton as follows. The main trondhjemite phase of the pluton was emplaced at about 3213 Ma as recorded by the zircon and apatite ages. It was later intruded by gabbroic dykes (the MGDs) at about 3179 Ma (hornblende $^{40}Ar/^{39}Ar$ age) and cooled through the hornblende blocking temperature (about 500 °C, McDougall & Harrison 1988). The difference between the zircon and hornblende ages indicates a cooling rate of about 6° Myr^{-1} from 700 °C to 500 °C. However, these data do not preclude an interpretation that the pluton cooled far more rapidly at about 3213 Ma (the zircon age) and then was reheated during the intrusion of the MGDs at 3179 Ma.

Muscovites from the pluton may have reached their blocking temperature (350 °C, McDougall & Harrison 1988) at about 3100 Ma, 80 Myr after the time of hornblende blocking. Mathematical modelling following Hall & York (1982) was performed on the age spectrum from the BT-1C muscovite, and the results are consistent with this sample cooling through its blocking temperature at about 3100 Ma, with some argon loss at about 3030 Ma. The difference between the hornblende and muscovite ages indicates a slower cooling rate, of about 2° Myr^{-1}, from 500 °C to 350 °C.

Modelling of the biotite age spectrum from sample BT-1C indicates that it has a blocking age of about 3030 Ma. The difference between the muscovite and biotite ages from sample BT-1C may be due either to slow cooling or to a resetting event at 3030 Ma which totally reset the biotite and only partially affected the muscovite. This thermal event may have been caused by the intrusion of massive sheet batholiths such as the Mpuluzi Batholith at about 3000 Ma (Anhaeusser & Robb 1983; Kröner *et al.* 1989).

The younger biotite ages from site BT-1F and the MGDs may reflect slow cooling following the Mpuluzi reheating event which reset the biotite and low-potassium muscovite argon clocks. Following this resetting event, the pluton cooled slowly or remained near the closure temperature of biotite for approximately another 100 Myr, as can be seen by the dispersion of biotite ages. Cooling rates for this temperature range (200–300 °C) would be much less than 1° Myr^{-1}.

Later DD emplacement is recorded by whole-rock ages of about 1900 Ma. Thus, apart from local resetting due to the dolerite dyke emplacement, it seems likely that the pluton cooled through the 600 °C to 400 °C temperature range near the time of hornblende blocking at 3179±18 Ma.

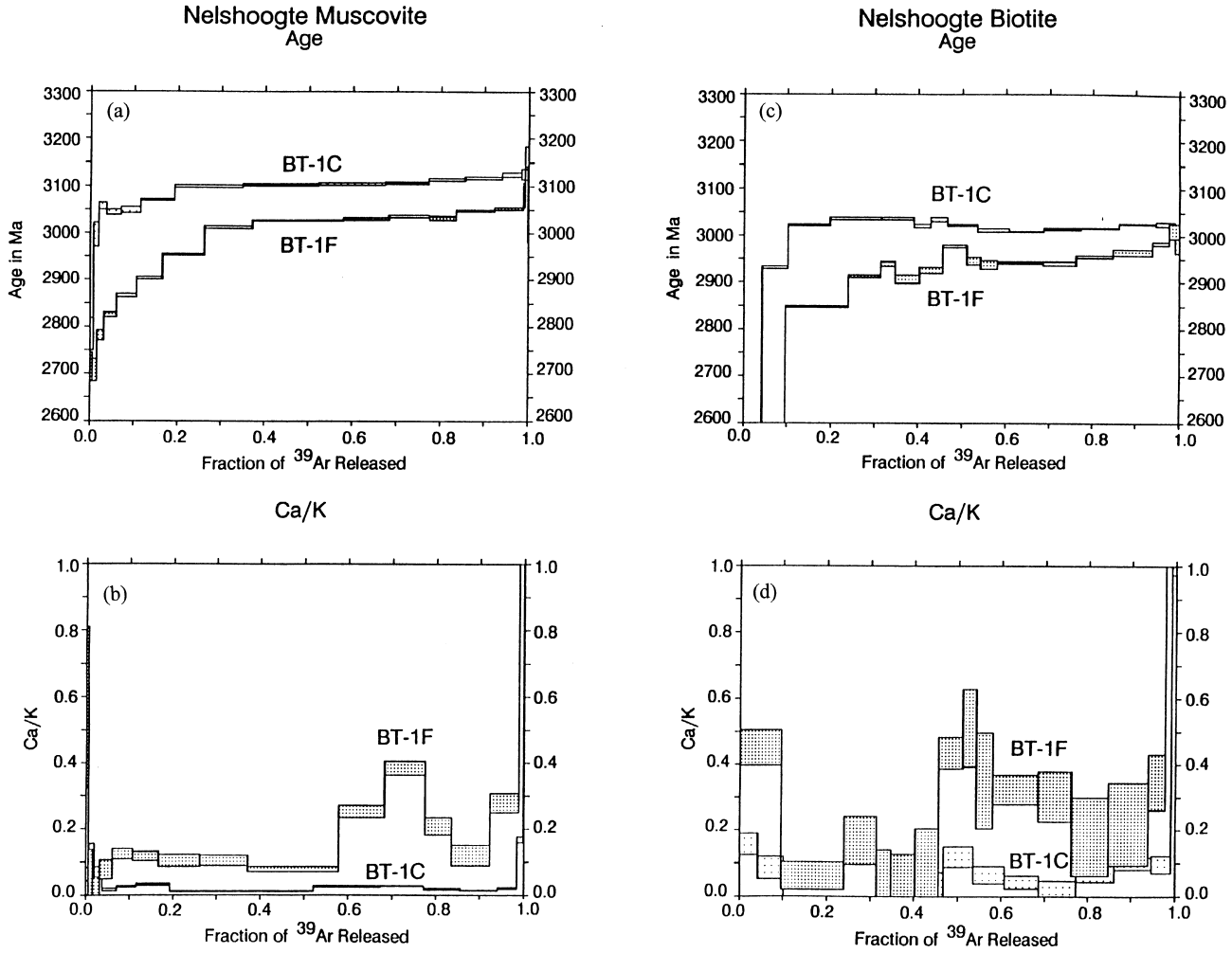


Figure 3. (a) $^{40}Ar/^{39}Ar$ age spectra and (b) Ca/K spectra for bulk-sample muscovites from samples BT-1C and BT-1F. (c), (d) Corresponding spectra for bulk-sample biotites from BT-1C (second run) and BT-1F. The heights of the shaded boxes represent $\pm 1\sigma$.

PALAEOMAGNETIC STUDY

31 sites (207 samples) from the trondhjemite of the NHP and five sites (31 samples) from cross-cutting dykes (Fig. 1) were sampled for palaeomagnetic study. Between four and 10 field-drilled cores were collected at each site. Sites in the pluton were generally chosen so as to be far from the pervasive dykes. However, owing to intermittent exposure, this could not always be guaranteed. In selected cases, sites were deliberately chosen so as to be in the reheating zone of the dykes. Most sites were river-cut outcrops, where the exposure was the freshest, although a few sites were located in rounded, domed outcrops away from rivers. We believe that sampling river exposures reduces the possibility of lightning strikes, because of their low elevation. The sites were located in the southern and central portions of the pluton because the northern part is not very well exposed, generally has a deeper weathering zone and is not readily accessible by vehicle. Three sites, numbers 18, 19, and 33, were located in highly eroded, intermittent-stream valleys where a very deep weathering zone made it impossible to obtain fresh specimens. Owing to questions about sample numbering in the field, samples from two sites (16 and 17) may have been misoriented. Rather than risk the chance that the wrong core orientation information was used to analyse the magnetic vectors, results from these sites are not included in the discussion.

One specimen from each core was stepwise demagnetized in successively higher alternating fields up to 80 mT in the cryogenic magnetometer at Stanford University, and stable demagnetization vectors were isolated using principal-component analysis (Kirschvink 1980). Selected specimens were thermally demagnetized in steps up to 700 °C for determination of magnetic carriers and unblocking temperatures.

Table 3 and Fig. 7 show the mean magnetic directions obtained from the trondhjemite. During demagnetization, most specimens showed stable univectorial behaviour, and generally one of two components is seen. It should be noted that in most sites there would often be one or two specimens which showed anomalous, yet stable, directions. Although we do not have an explanation for this dispersion, a similar behaviour was observed in the Kaap Valley Pluton (Layer, Kröner & McWilliams 1996). Evans (1967) reported similar behaviour in the Gaborone Granite of Botswana. It is possible that this is due to the slow cooling of the pluton or to heterogeneities in the mineralogy of the pluton. Nevertheless, two dominant

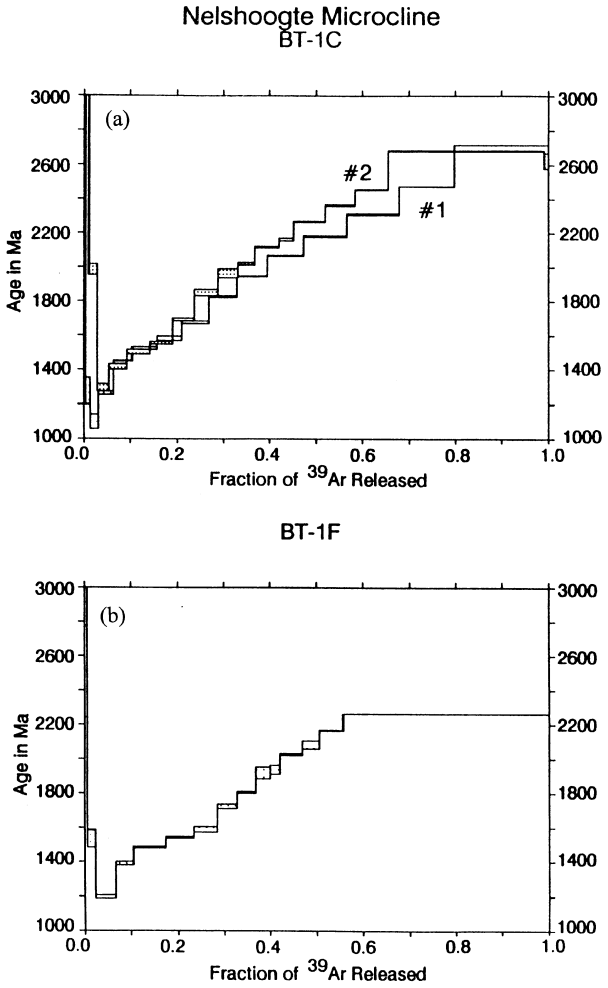


Figure 4. $^{40}Ar/^{39}Ar$ age spectra from microcline from (a) sample BT-1C (2 runs) and (b) sample BT-1F (1 run). The heights of the shaded boxes represent $\pm 1\sigma$.

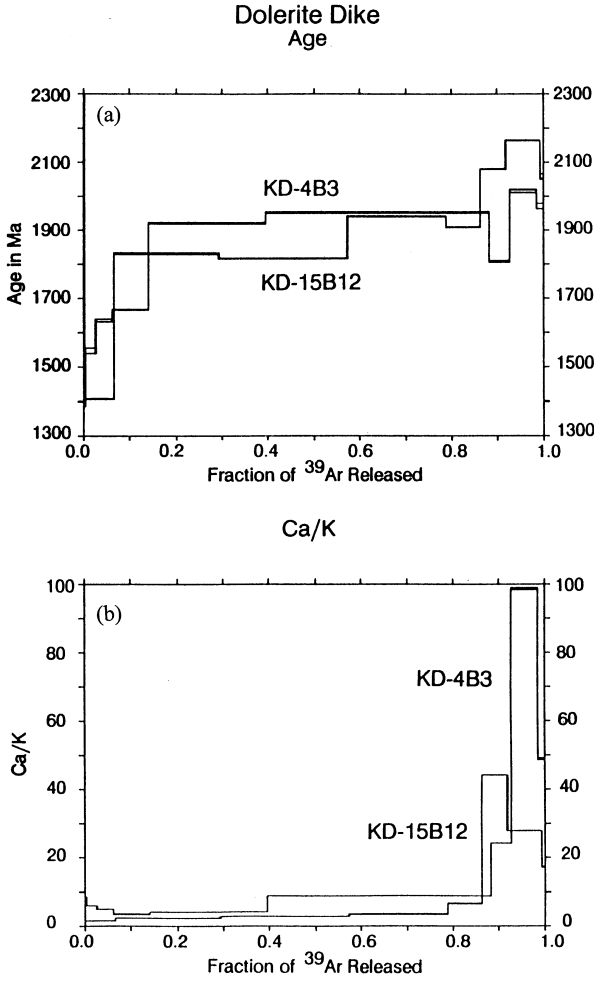


Figure 5. (a) $^{40}Ar/^{39}Ar$ age and (b) Ca/K spectra from dolerite dyke whole-rock samples KD-4B3 and KD-15B12. The heights (very small) of the shaded boxes represent $\pm 1\sigma$.

components were isolated from specimens at a majority of sites throughout the pluton. The first of these directions, direction A, is seen in 17 sites in the trondhjemite. 11 sites show a northwesterly directed component (A+, Table 4), whereas six sites showed a much more scattered southeasterly direction (A−). The palaeomagnetic poles calculated from the site mean VGPs of these two directions are 167° apart; however, given the large A_{95} on the A− direction, the two magnetic directions are antipodal, and the mean palaeomagnetic pole for all 17 sites is 309.8°E, 17.6°N, $K=16$, $A_{95}=9.4°$ (Table 4).

Thermal demagnetization of specimens showing the A direction showed that this component is stable and has a distributed unblocking-temperature range reaching a maximum of 560–580 °C, indicating titanomagnetite as the primary magnetic remanence carrier. The A component is always the highest-coercivity or highest-blocking-temperature component present, and in one specimen, both the A+ and A− components were seen during alternating-field (AF) demagnetization.

The second component, B, is seen in specimens from 11 sites. AF and thermal-demagnetization experiments show behaviour similar to that seen in specimens showing the A component, and two approximately antipodal directions could be isolated. A steep, positive-inclination component (B+) is present in six sites, while a steep, negative-inclination component (B−) is found in five sites (Tables 3 and 4, Fig. 7). The palaeomagnetic poles calculated from the site mean VGPs of these two groups are 143° apart. Such a departure from antipodal directions may imply that the two polarities were acquired at different times. The large A_{95} values make it difficult to confirm that the two polarities are from different populations. In the following discussions of the B component, therefore, we will not combine the B+ and B− palaeomagnetic poles into an averaged palaeomagnetic pole.

In one specimen, the B− component is present as an overprint on the A+ component. The B+ and B− palaeomagnetic directions have steep inclinations, and the corresponding palaeomagnetic poles are similar to others dated at about 2000 Ma, such as that from the Bushveld Complex (Hattingh 1983), the Witwatersrand Supergroup overprint (Layer *et al.* 1988b) and the Vredefort Dome (Hargraves 1970) (Table 4). Examination of the site directions of the B+ and B− components (Fig. 7, Table 4) and the resultant VGPs indicates that the distributions may not be Fisherian, and seem to be 'smeared' along a southwest to northeast path. It may be that each site mean B direction represents a slightly different time along an apparent polar wander path.

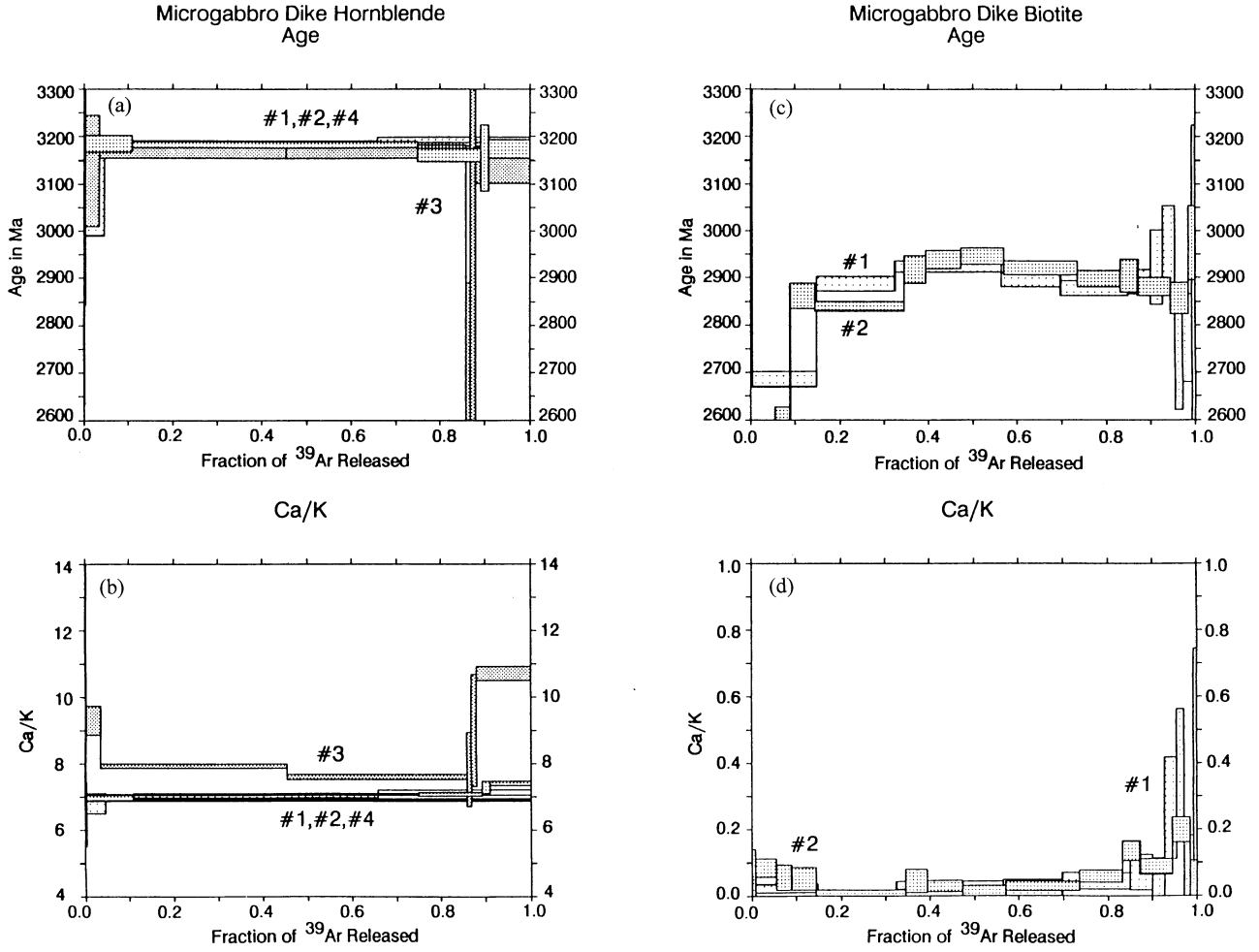


Figure 6. (a) $^{40}Ar/^{39}Ar$ age and (b) Ca/K spectra from four single hornblende grains from site 21. (c), (d) Corresponding results from two single biotite grains from the same site. The heights of the shaded boxes represent $\pm 1\sigma$. Shading patterns are used to distinguish between the different samples (see Appendix A for analytical data).

Two dyke localities in the NHP were sampled. Both outcrops are exposed in the Komati River and are located 3 km apart. The first locality is in a 1.5 m wide MGD (site 21). The second locality is made up of several sites (numbers 4, 10, 11, 23 and 24) in a small dyke swarm of varying composition, although the relationship between the coarser-grained MGDs and the fine-grained DDs was not noted in the field.

Two magnetic directions similar to those found in the trondhjemite are seen in the dykes, and correlate with dyke lithology. The MGD, which has an Archaean $^{40}Ar/^{39}Ar$ age, shows only the A direction, and samples from the MGD show demagnetization characteristics similar to those from the trondhjemite specimens (unblocking temperatures from 560° to 600 °C). The well-defined palaeomagnetic direction (Fig. 7) and VGP (20.6°S, 134.6°E) from site 21 show the A− direction, and the VGP is 175° away from the mean A palaeomagnetic pole determined from the trondhjemite samples, while the less well-defined A+ VGP (34.2°N, 331.4°E) seen in site 23 is 26° away ($A_{95}=31°$). This implies that the magnetizations of the trondhjemite and the MGD were probably acquired at the same time.

The DDs, which have Proterozoic ages, show both the B+ and B− directions. The DD specimens show a very different behaviour, with two discrete unblocking temperatures at 400 °C and 580 °C. The B+ palaeomagnetic pole from sites 4 and 11 as calculated from six specimen VGPs (19.6°S, 17.5°E, $K=11$, $A_{95}=21°$) is only 18° away from the B+ pole from the pluton, while the B− palaeomagnetic pole from site 10 calculated from four specimen VGPs (14.5°S, 216.7°E, $K=63$, $A_{95}=12°$) is 27° away from the B− pole from the pluton (Table 4). As was seen in the trondhjemite, these two poles are not antipodal (they are 140° apart).

The direction at site 24 is shallower than the B+ direction found elsewhere (Fig. 7). At this site, the dyke is very thin (10 cm wide), and secular variation was probably not averaged during dyke cooling. This is compatible with the high k-value of 228 (Table 3). A second possible explanation of the directions from this site is that this small dyke was intruded relatively late (it is observed to cross-cut all other dykes) and has a corresponding palaeomagnetic pole further down the apparent polar wander path for the lower Proterozoic.

DISCUSSION OF THE PALAEOMAGNETIC STUDY

The palaeomagnetic study shows that a stable, consistently shallow northeast direction (A+) is present in 11 sites in the Nelshoogte Pluton. The between-site dispersion is moderately

Table 3. Site mean palaeomagnetic data.

Site	*N*	*n*	*D*	*I*	*k*	α_{95}	Dir.	Lat.	Lon.
Trondhjemite pluton									
1	4	3	282.8	2.7	9	49.3	A+	10.9	307.5
2	4	4	251.0	41.0	22	20.0	U	−26.3	315.3
3	4	3	307.3	25.3	9	42.8	A+	25.5	331.6
5	4	4	115.6	−29.7	15	24.9	A−	−14.7	146.9
6	4	3	286.6	−38.4	19	28.8	A+	23.6	287.0
		1	195.0	−75.0			B−	−1.5	217.6
7	4	4	285.3	21.7	5	45.0	A+	8.5	317.6
8	4	0	Multiple overprints, no stable vectors isolated						
9	4	2	82.9	3.6			A−	5.6	115.9
		2	10.0	−80.0			B−	45.0	205.9
12	4	4	303.2	16.7	26	18.3	A+	25.0	324.7
13	4	2	201.1	−63.4			B−	−16.6	226.0
		1	123.2	86.0			B+	−30.1	38.3
14	4	3	102.9	11.2	24	25.8	A−	−14.1	121.2
15	4	1	295.0	5.0			A+	21.2	314.5
16	4	0	Field labelling error						
17	4	0	Field labelling error						
18	6	0	Altered/weathered, no stable vectors isolated						
19	6	0	Altered/weathered, no stable vectors isolated						
20	8	6	288.0	63.7	7	27.9	U	−6.6	348.3
22	10	5	233.8	62.1	26	12.0	B+	−43.3	336.8
25	10	4	286.8	23.8	8	36.0	A+	9.2	319.3
		3	208.1	66.6	24	28.0	B+	−58.2	354.8
26	10	10	280.2	27.2	3	32.0	A+	2.6	318.0
27	10	7	150.1	30.3	6	27.0	A−	−60.6	133.6
28	10	4	293.6	3.6	4	53.0	A+	20.3	313.2
		3	212.5	−71.7	11	39.9	B−	−3.2	228.0
29	10	5	80.0	26.0	9	27.8	A−	2.8	104.0
		2	245.0	87.0			B+	−28.2	24.5
30	10	5	342.2	47.4	14	21.0	U	33.0	12.0
		2	326.6	−83.4			B−	36.4	219.6
31	10	3	94.4	21.5	8	46.0	A−	−8.7	112.5
32	10	5	314.7	−2.9	14	20.9	A+	40.1	322.5
		3	250.3	−43.2	64	15.5	U	−5.1	269.5
		1	347.9	77.3			B+	−2.1	25.8
33	5	0	Altered/weathered, no stable vectors isolated						
34	5	2	287.1	−13.0			A+	18.2	302.0
		2	204.5	74.1			B+	−51.6	11.3
35	10	7	222.4	34.6	117	5.6	U	−50.4	299.6
Microgabbro dykes									
21	8	7	114.6	−5.6	17	15.0	A−	−20.6	134.6
23	8	4	312.4	15.2	10	31.0	A+	34.2	331.4
Dolerite dykes									
4	4	4	265.5	81.8	59	12.1	B+	−26.0	12.7
10	4	4	188.7	−65.5	148	7.6	B−	−14.5	216.7
11	4	2	344.1	72.7			B+	−6.2	26.3
4+11	8	6	295.7	75.9	37	11.2	B+	−18.1	17.5
24	4	3	7.1	40.5	228	8.2	B+?	40.5	39.2

Site: sampling site; *N*: number of specimens per site; *n*: number of specimens used in calculation of mean direction; *D*, *I*, Lat. Lon.: site mean declination, inclination, VGP latitude (°N) and VGP longitude (°E); *k*: Fisherian precision parameter; α_{95}: radius of a circle of 95 per cent confidence about the mean direction; Dir.: palaeomagnetic direction classification (A, direction A was used to calculate the mean A pole; B, direction was B used to calculate the mean B pole; U, unknown direction; + and − refer to the polarity of the direction).

low ($K = 28$ for the palaeomagnetic pole), and the sites are located over a wide area in the pluton. A less well-defined, antipodal population was isolated at six sites. These directions are also seen in the MGD within the pluton. A palaeomagnetic pole calculated from the 17 pluton sites (18°N, 310°E, $K = 16$, $A_{95} =$ 9°, shown in Fig. 8) implies an equatorial palaeolatitude of 0°. Inclusion of the MGD results does not significantly change this result. On the basis of the similarity between the MGD VGP and the A pole, and the unblocking temperatures seen in the A component, we suggest that the age of magnetization of the

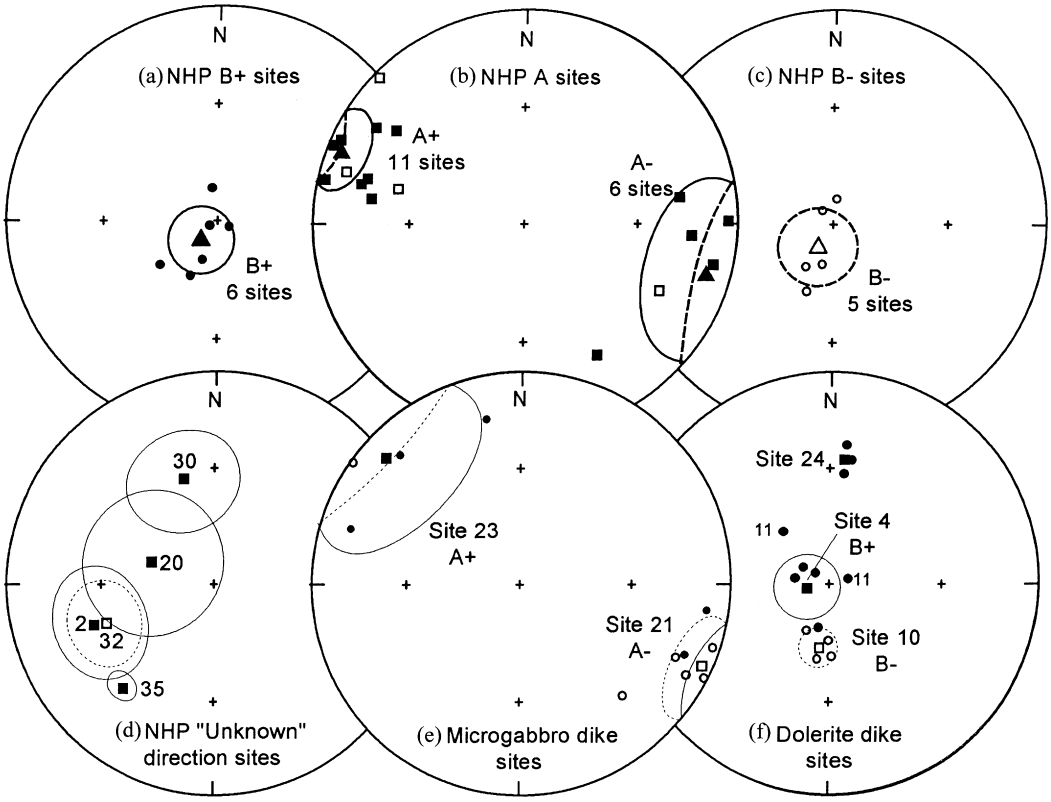


Figure 7. Equal-area stereographic projections of palaeomagnetic data. (a), (b), (c), (d) Site and study mean magnetic directions from the trondhjemite palaeomagnetic sites. The filled and open squares are the site mean directions projected on the lower and upper hemispheres, respectively. The large filled and open triangles represent the mean of each grouping projected on the lower and upper hemispheres, respectively. The study mean 95 per cent confidence limits are shown by solid (lower hemisphere) and dashed (upper hemisphere) ellipses. In (a), (b) and (c), circles of confidence for individual sites are omitted for clarity. (e), (f) Individual specimen directions and site means from the microgabbro and dolerite dyke sites. The filled and open circles are the individual specimen directions projected on the lower and upper hemispheres, respectively. The filled and open squares are the site means projected on the lower and upper hemispheres, respectively. The specimen directions in (f) labelled 11 are from site 11.

trondhjemite is best represented by the age of hornblende from the MGD (3179 ± 18 Ma). The ages from the trondhjemite are consistent with the hornblende age being the blocking age, with zircon ($T_B = 700\,^{\circ}C$ at 3213 Ma) being older and muscovite ($T_B = 350\,^{\circ}C$ at 3107 Ma) being younger. It should be noted that the A direction seen in the Nelshoogte pluton is also seen as an overprint in samples from the adjacent Kaap Valley pluton, identified as the C1 direction in Layer *et al.* (1996).

The second, less well-defined, stable direction (B), with approximately antipodal components, is seen in the trondhjemite and the DD samples. This magnetic direction is seen as an overprint on (it is younger than) the A direction in the NHP. The results from the dolerite dykes support the idea that the B magnetization was acquired at the time of dolerite intrusion. The similarity of the B pole to other early-Proterozoic poles and the whole-rock $^{40}Ar/^{39}Ar$ ages from the dykes are consistent with the idea that the B magnetization is post-Archaean (Table 4). The departure from antipodality of the B+ and B− poles may reflect the fact that dyke intrusion at this time was a protracted event.

Van der Voo (1990) has developed a seven-criterion *Q*-factor to assess palaeomagnetic poles. We assign the Nelshoogte A pole a *Q*-factor score of 6 on the basis of (1) a well-defined magnetization age; (2) a sufficient number of samples ($N > 24$, $k > 10$ and $\alpha_{95} < 16$); (3) stepwise demagnetization and vector identification; (4) partial overprinting of the A magnetic direction by the Proterozoic B direction, indicating that the magnetization pre-dates approximately 2000 Ma; (5) antipodal reversals in the pluton; and (6) the lack of similarity of the Nelshoogte pole to other Archaean and Proterozoic poles. The seventh criterion is that of structural control and tectonic coherence. With a pluton it is difficult to determine whether or not there has been significant rotation and we do not feel confident in assigning this criterion to the Nelshoogte A direction. However, the presence of the B direction, which is widespread in the Kaapvaal Craton, seems to indicate that the pluton has not rotated relative to the craton since approximately 2000 Ma. Also, the similarity of the A direction to that of the C1 overprint in the Kaap Valley pluton seems to imply that at least these two bodies acted as a single structural block since approximately 3180 Ma, so it is possible that the A direction does indeed reflect the true palaeolatitude of the Kaapvaal Craton at that time.

CONCLUSIONS

The new $^{207}Pb/^{206}Pb$ and $^{40}Ar/^{39}Ar$ ages are consistent with previous results from the Nelshoogte Pluton, but the new ages are far more precise. The range in ages spans over 200 Myr and indicates that this pluton underwent slow cooling from 700 °C to 200 °C and also experienced resetting events later in its history. The combination of dating by two different methods

Table 4. Summary of palaeomagnetic data.

Pole	*N*	*D*	*I*	*k*	a_{95}	Lat.	Lon.	*K*	A_{95}	Reference
Nelshoogte trondhjemite pluton										
A+	11	293.1	7.0	13	13	19.0	314.3	28	9	This study
A−	6	104.0	11.8	7	28	14.7	121.3	9	24	This study
A	17	285.1	0.6	9	13	17.6	309.8	16	9	This study
B+	6	224.7	80.0	28	13	37.6	15.0	9	23	This study
B−	5	212.9	−80.6	27	15	11.9	220.3	9	28	This study
Other palaeomagnetic poles: Archaean										
Komati Formation (3470±20 Ma)						−11	21	60	15	Hale & Dunlop (1984)
Kaap Valley Pluton A (3214±4 Ma)						51	170	48	5	Layer *et al.* (1996)
Usushwana Complex (2875±40 Ma)						9	347	22	8	Layer *et al.* (1988a)
Mbabane Pluton (2687±6 Ma)						20	106	67	9	Layer *et al.* (1989)
Modipe Gabbro (2575±470 Ma)						−33	31		11	Evans & McElhinny (1966)
Other palaeomagnetic poles: Palaeoproterozoic										
Great Dyke (2460±16 Ma)						22	59		8	McElhinny & Gough (1963); Wilson, Jones & Kramers (1987)
Gaborone Granite (2330±45 Ma)						34	103		16	Evans (1967); Harding, Crockett & Snelling (1974)
Ongeluk Lavas (Transvaal Group) (2222±13 Ma)						1	281	38	5	Evans *et al.* (1997)
Bushveld Complex (2050±12 Ma)						16	30	21	4	Hattingh (1983)
Vredefort Dome (1950±60 Ma)						28	34	21	11	Hargraves (1970)
Witwatersrand (1945±40 Ma)						19	46	22	8	Layer *et al.* (1988b)

Pole: palaeomagnetic pole; *N*: number of sites used in the calculation of the pole; *D*, *I*, Lat. Lon.: mean declination, inclination, latitude (°N) and longitude (°E) of the palaeomagnetic pole; *k*, a_{95}: Fisherian precision parameter and radius of circle of 95 per cent confidence about the mean direction; *K*, A_{95}: Fisherian precision parameter and radius of circle of 95 per cent confidence about the mean pole.

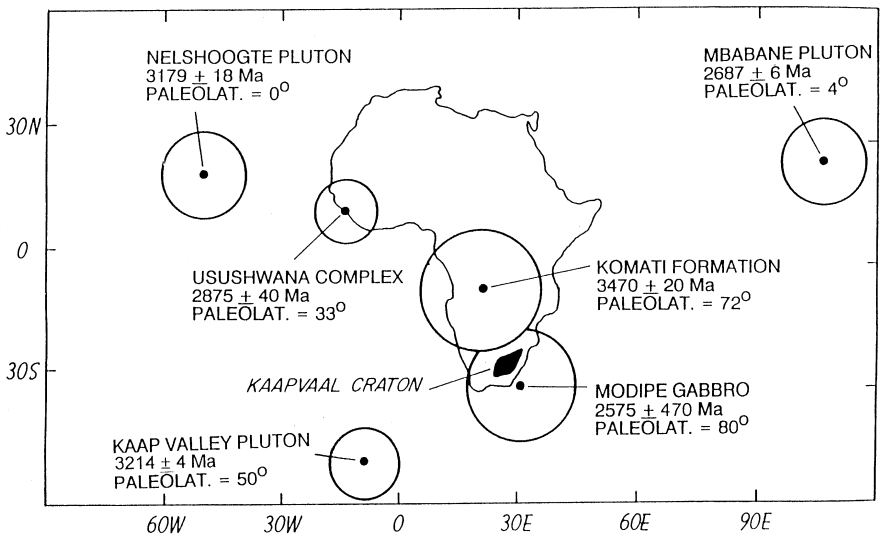


Figure 8. Mercator-projection map showing the six Archaean (pre-2500 Ma) palaeomagnetic poles for the Kaapvaal Craton. With each pole is the age associated with the magnetism and the palaeolatitude of the craton relative to that pole. (See Table 4 for references).

and using several minerals allows for precise determination of the thermal history of the pluton.

The palaeomagnetic study of the NHP provides a well-documented, precisely dated, equatorial palaeomagnetic pole at 3179±18 Ma. This pole is significantly different from others thus far obtained for the Archaean in southern Africa (Fig. 8). When these data are combined with other Archaean palaeomagnetic poles, a picture of significant apparent polar wander in the Archaean for the Kaapvaal Craton emerges as suggested by Kröner & Layer (1992), contradicting Hale's (1987) hypothesis. With only six Archaean poles spanning almost 1000 million years, meaningful estimates of plate velocities cannot be calculated. However, this study does provide evidence for significant cratonic movement (as seen by apparent polar wander) prior to 2500 Ma, extending Evans's (1967) observation of early Proterozoic apparent polar wander back into the middle Archaean. In addition, the approximate antipodal nature of the site mean magnetic directions as was seen in the 3214 Ma Kaap

Valley pluton (Layer *et al.* 1996) implies that the Earth's magnetic field was undergoing reversals in the Archaean.

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APPENDIX A: GEOCHRONOLOGIC DATA

The following tables are the data for the bulk sample and single grain $^{40}Ar/^{39}Ar$ experiments.

For bulk samples, the mass is given in grams and the volume of ^{39}Ar given is in m lg^{-1}, for single grains, the volume of ^{39}Ar given is the total volume measured in ml.

The error of the integrated age includes the analytical error of *J*. All errors are $\pm 1\sigma$.

Measured ratios are corrected for decay of ^{37}Ar and ^{39}Ar. 37Ca/39K: (^{37}Ar from calcium)/(^{39}Ar from potassium)
40*/39K: (radiogenic ^{40}Ar)/(^{39}Ar from potassium)
Constants used:

$\lambda_{e+e'} = 0.581 \times 10^{-10} a^{-1}$
$\lambda_{\beta} = 4.962 \times 10^{-10} a^{-1}$
$^{40}K/K = 1.167 \times 10^{-4}$
$(^{40}Ar/^{39}Ar)K = 2.87 \times 10^{-2}$
$(^{37}Ar/^{39}Ar)_{Ca} = 1536.1$
$(^{36}Ar/^{39}Ar)_{Ca} = 0.390169$

MLM 79 MUSCOVITE BT-1C
Mass=0.0072 g

Weighted average of J from standards = 0.036455 +/- 0.000662

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-6 ml/g	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
500	0.0085	115.524	0.035	0.049	0.181	12.630	0.035	0.010	100.936	2.400	2784.1	33.7
550	0.0192	125.174	0.067	0.028	0.228	6.594	0.067	0.018	116.924	2.126	2995.9	26.6
600	0.0381	124.114	0.038	0.008	0.401	1.793	0.038	0.009	121.891	0.567	3056.9	6.8
650	0.0713	122.958	0.009	0.007	0.708	1.721	0.009	0.003	120.843	0.448	3044.2	5.4
700	0.1147	123.368	0.014	0.007	0.924	1.715	0.014	0.002	121.254	0.561	3049.2	6.8
750	0.1929	123.907	0.017	0.003	1.664	0.732	0.017	0.001	123.000	0.230	3070.2	2.8
800	0.3492	125.962	0.005	0.002	3.328	0.438	0.005	0.001	125.411	0.207	3098.9	2.4
850	0.5230	126.230	0.006	0.002	3.699	0.416	0.006	0.001	125.706	0.141	3102.4	1.7
900	0.6740	126.578	0.014	0.003	3.217	0.587	0.014	0.001	125.836	0.218	3103.9	2.6
925	0.7740	127.321	0.015	0.004	2.127	1.019	0.015	0.001	126.025	0.207	3106.1	2.4
950	0.8564	128.378	0.010	0.006	1.755	1.339	0.010	0.002	126.659	0.267	3113.6	3.1
1000	0.9390	129.476	0.007	0.009	1.758	1.952	0.007	0.001	126.949	0.262	3117.0	3.1
1025	0.9836	131.659	0.011	0.014	0.950	3.061	0.011	0.002	127.629	0.343	3124.9	4.0
1050	0.9988	144.185	0.092	0.056	0.324	11.528	0.092	0.005	127.572	0.954	3124.2	11.1
1100	1.0000	411.526	3.323	0.978	0.026	70.132	3.331	0.140	123.182	7.603	3072.4	91.1
Integrated		127.097	0.017	0.007	21.290	1.603	0.017	0.000	125.060	0.069	3094.8	26.9

MLM 74 MUSCOVITE BT-1F S.A.
Mass=0.0098 g

Weighted average of J from standards = 0.027660 +/- 0.000074

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-6 ml/g	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
540	0.0064	145.786	0.310	0.065	0.069	13.206	0.310	0.132	126.545	2.725	2713.6	30.2
582	0.0148	135.549	-0.054	0.032	0.090	6.998	-0.054	0.129	126.045	2.126	2708.0	23.6
625	0.0316	135.741	-0.027	0.010	0.179	2.174	-0.027	0.056	132.772	1.089	2781.3	11.6
670	0.0606	138.776	0.042	0.006	0.310	1.366	0.042	0.016	136.870	0.475	2824.6	4.9
715	0.1074	142.147	0.068	0.004	0.500	0.885	0.068	0.009	140.880	0.326	2865.9	3.3
757	0.1668	145.944	0.064	0.005	0.634	0.935	0.064	0.008	144.571	0.259	2903.1	2.6
805	0.2631	150.444	0.057	0.002	1.028	0.424	0.057	0.010	149.796	0.248	2954.6	2.4
852	0.3714	156.557	0.057	0.002	1.156	0.411	0.057	0.009	155.904	0.220	3012.9	2.1
900	0.5771	158.492	0.042	0.003	2.197	0.648	0.042	0.005	157.453	0.146	3027.4	1.4
925	0.6830	159.327	0.138	0.005	1.131	0.887	0.138	0.010	157.912	0.303	3031.6	2.8
950	0.7750	161.537	0.210	0.011	0.982	1.945	0.210	0.012	158.401	0.237	3036.2	2.2
975	0.8365	161.470	0.113	0.012	0.657	2.143	0.113	0.014	158.006	0.457	3032.5	4.2
1000	0.9214	164.708	0.065	0.017	0.907	3.099	0.065	0.017	159.595	0.239	3047.2	2.2
1050	0.9889	166.614	0.152	0.021	0.720	3.804	0.152	0.015	160.278	0.334	3053.5	3.1
1102	0.9944	258.555	5.829	0.326	0.059	37.099	5.852	0.141	163.243	2.762	3080.5	25.0
1600	1.0000	199.472	5.472	0.095	0.060	13.806	5.491	0.122	172.534	2.384	3162.7	20.6
Integrated		157.182	0.149	0.010	10.678	1.902	0.149	0.004	154.192	0.074	2996.7	4.0

MLM 45 BIOTITE BT-1C FIRST RUN
Mass=0.0070 g

Weighted average of J from standards = 0.034711 +/- 0.000202

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-6 ml/g	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
493	0.0184	53.050	0.125	0.018	0.295	10.014	0.125	0.004	47.727	0.955	1762.7	22.5
542	0.0543	97.685	0.026	0.008	0.579	2.320	0.026	0.003	95.405	0.573	2636.3	8.3
580	0.1099	121.867	0.009	0.003	0.893	0.755	0.009	0.002	120.932	0.407	2973.5	4.9
642	0.2140	125.981	0.002	0.004	1.675	0.977	0.002	0.002	124.734	0.190	3018.7	2.2
690	0.3322	126.867	0.002	0.002	1.901	0.501	0.002	0.001	126.216	0.139	3036.1	1.6
730	0.4020	126.741	0.004	0.003	1.122	0.749	0.004	0.002	125.776	0.451	3030.9	5.3
785	0.4418	126.198	0.018	0.006	0.640	1.492	0.018	0.003	124.301	0.407	3013.6	4.8
830	0.4855	128.561	0.034	0.007	0.703	1.576	0.034	0.002	126.523	0.413	3039.6	4.8
875	0.5571	125.382	0.022	0.004	1.152	1.050	0.022	0.003	124.053	0.424	3010.7	5.0
900	0.6156	124.886	0.016	0.004	0.940	1.023	0.016	0.003	123.594	0.434	3005.3	5.1
925	0.7084	125.692	0.013	0.004	1.494	0.943	0.013	0.001	124.492	0.206	3015.9	2.4
945	0.7915	126.196	0.020	0.006	1.336	1.399	0.020	0.002	124.417	0.315	3015.0	3.7
980	0.8714	126.676	0.019	0.007	1.285	1.625	0.019	0.001	124.604	0.180	3017.2	2.1
1017	0.9527	127.666	0.024	0.008	1.308	1.826	0.024	0.003	125.322	0.195	3025.6	2.3
1065	0.9957	133.731	0.041	0.027	0.691	6.059	0.041	0.002	125.617	0.301	3029.1	3.5
1600	1.0000	174.985	1.524	0.127	0.070	21.415	1.526	0.072	137.637	5.162	3164.3	56.0
Integrated		124.296	0.024	0.007	16.086	1.572	0.024	0.001	122.329	0.082	2990.3	8.6

MLM 59 BIOTITE BT-1C 2ND RUN
Mass=0.0089 g

Weighted average of J from standards = 0.034711 +/- 0.000202

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-6 ml/g	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
542	0.0418	81.162	0.087	0.009	0.604	3.109	0.087	0.018	78.628	0.484	2374.6	8.1
580	0.1029	118.792	0.047	0.004	0.883	1.109	0.047	0.019	117.462	0.213	2931.2	2.6
642	0.1981	125.786	-0.017	0.003	1.377	0.615	-0.017	0.013	124.996	0.200	3021.8	2.3
690	0.3165	126.724	-0.013	0.002	1.710	0.378	-0.013	0.008	126.228	0.149	3036.2	1.7
730	0.3899	127.157	0.034	0.003	1.062	0.759	0.034	0.006	126.178	0.159	3035.6	1.9
785	0.4290	126.985	0.011	0.007	0.565	1.702	0.011	0.014	124.809	0.325	3019.6	3.8
830	0.4662	129.339	0.009	0.011	0.538	2.568	0.009	0.030	126.003	0.308	3033.6	3.6
875	0.5362	127.833	0.065	0.010	1.011	2.252	0.065	0.018	124.945	0.162	3021.2	1.9
900	0.6060	127.224	0.036	0.011	1.010	2.464	0.036	0.015	124.076	0.220	3011.0	2.6
925	0.6875	127.793	0.024	0.013	1.178	3.052	0.024	0.011	123.879	0.136	3008.7	1.6
945	0.7731	128.931	0.003	0.015	1.237	3.531	0.003	0.022	124.363	0.211	3014.4	2.5
980	0.8589	130.061	0.049	0.019	1.240	4.296	0.049	0.026	124.463	0.111	3015.6	1.3
1017	0.9459	132.658	0.054	0.025	1.257	5.584	0.054	0.010	125.240	0.137	3024.7	1.6
1065	0.9919	145.144	0.052	0.067	0.666	13.687	0.052	0.015	125.269	0.365	3025.0	4.3
1162	1.0000	281.364	0.726	0.540	0.117	56.647	0.726	0.194	122.030	1.870	2986.7	22.4
Integrated		127.674	0.033	0.017	14.454	3.955	0.033	0.005	122.612	0.053	2993.6	8.5

MLM 77 BIOTITE BT-1F
Mass=0.0115 g

Weighted average of J from standards = 0.027660 +/- 0.000074

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-6 ml/g	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
525	0.0966	112.208	0.246	0.134	0.730	35.159	0.246	0.029	72.758	0.423	1989.5	7.0
600	0.2390	161.608	0.034	0.076	1.075	13.942	0.034	0.023	139.067	0.222	2847.3	2.3
650	0.3132	164.118	0.093	0.063	0.560	11.287	0.093	0.040	145.590	0.326	2913.3	3.2
700	0.3469	162.153	0.022	0.047	0.254	8.617	0.022	0.054	148.169	0.446	2938.7	4.4
750	0.4016	163.451	0.019	0.063	0.413	11.302	0.019	0.051	144.966	0.888	2907.1	8.8
800	0.4557	166.868	0.006	0.068	0.409	11.955	0.006	0.107	146.905	0.567	2926.3	5.6
850	0.5101	163.709	0.238	0.039	0.411	7.039	0.238	0.026	152.194	0.284	2977.7	2.7
875	0.5426	159.808	0.279	0.037	0.246	6.825	0.279	0.064	148.913	0.865	2946.0	8.4
900	0.5793	158.170	0.192	0.034	0.277	6.417	0.192	0.079	148.025	0.874	2937.3	8.6
925	0.6848	155.907	0.177	0.025	0.797	4.785	0.177	0.024	148.448	0.181	2941.4	1.8
950	0.7614	155.286	0.165	0.024	0.578	4.508	0.165	0.041	148.287	0.482	2939.9	4.7
975	0.8474	155.475	0.099	0.019	0.650	3.654	0.099	0.065	149.789	0.269	2954.5	2.6
1000	0.9373	157.610	0.120	0.023	0.679	4.339	0.120	0.068	150.769	0.551	2964.0	5.3
1050	0.9780	162.897	0.189	0.034	0.308	6.201	0.189	0.046	152.800	0.372	2983.5	3.6
1100	1.0000	189.111	1.709	0.114	0.166	17.808	1.711	0.227	155.594	1.629	3010.0	15.3
Integrated		156.040	0.163	0.054	7.553	10.181	0.163	0.014	140.154	0.111	2858.5	4.0

MLM 40 MICROCLINE BT-1C 1ST RUN
Mass=0.0008 g

Weighted average of J from standards = 0.037962 +/- 0.000313

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-6 ml/g	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
550	0.0039	127.514	0.192	0.062	0.441	14.387	0.192	0.110	109.169	3.360	2954.9	44.7
600	0.0132	40.866	-0.095	0.046	1.056	33.566	-0.095	0.048	27.137	2.306	1277.5	77.8
650	0.0315	27.521	-0.106	0.018	2.083	19.640	-0.106	0.028	22.102	1.065	1099.1	39.7
700	0.0635	29.871	0.033	0.010	3.632	10.139	0.033	0.034	26.829	0.216	1267.1	7.3
750	0.1051	34.654	0.020	0.007	4.726	5.789	0.020	0.013	32.634	0.175	1454.0	5.4
800	0.1576	36.022	0.006	0.003	5.959	2.658	0.006	0.007	35.050	0.213	1526.5	6.3
850	0.2120	38.737	0.016	0.006	6.175	4.454	0.016	0.025	36.997	0.304	1582.8	8.7
900	0.2694	41.999	0.013	0.005	6.526	3.642	0.013	0.004	40.455	0.308	1678.7	8.3
950	0.3280	47.769	0.005	0.005	6.645	3.138	0.005	0.010	46.255	0.153	1828.9	3.8
1000	0.3958	52.613	0.014	0.005	7.708	2.682	0.014	0.006	51.187	0.128	1947.5	3.0
1050	0.4727	58.456	0.006	0.007	8.726	3.422	0.006	0.011	56.441	0.140	2065.8	3.0
1100	0.5670	63.737	0.014	0.006	10.705	2.880	0.014	0.005	61.887	0.086	2180.7	1.8
1150	0.6793	70.617	-0.006	0.007	12.753	2.898	-0.006	0.003	68.554	0.106	2312.2	2.0
1200	0.7977	80.212	0.003	0.009	13.446	3.384	0.003	0.009	77.483	0.112	2474.4	1.9
1600	1.0000	93.581	-0.009	0.004	22.977	1.368	-0.009	0.004	92.285	0.084	2714.8	1.3
Integrated		63.418	0.003	0.007	113.559	3.202	0.003	0.003	61.372	0.051	2170.2	10.5

MLM 58 MICROCLINE BT-1C 2ND RUN
Mass=0.0027 g

Weighted average of J from standards = 0.034711 +/- 0.000202

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-6 ml/g	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
580	0.0105	360.361	0.045	0.044	0.282	3.599	0.045	0.159	347.385	3.885	4635.4	18.6
642	0.0288	60.561	0.197	0.009	0.491	4.593	0.197	0.142	57.772	1.468	1985.2	30.6
690	0.0546	34.193	-0.048	0.013	0.691	11.157	-0.048	0.103	30.363	0.794	1298.5	24.2
730	0.0942	38.336	0.043	0.013	1.062	10.091	0.043	0.044	34.455	0.609	1419.1	17.4
785	0.1412	38.718	-0.019	0.004	1.259	2.889	-0.019	0.034	37.584	0.592	1506.2	16.1
830	0.1905	42.262	0.018	0.009	1.323	6.332	0.018	0.028	39.571	0.319	1559.4	8.4
875	0.2388	48.043	0.075	0.011	1.295	6.523	0.075	0.027	44.897	0.277	1694.7	6.8
925	0.2894	53.692	0.108	0.007	1.356	3.786	0.108	0.062	51.648	0.753	1852.8	16.9
945	0.3318	59.019	-0.066	0.008	1.136	3.877	-0.066	0.124	56.713	1.276	1963.0	26.9
980	0.3687	62.980	0.035	0.012	0.990	5.430	0.035	0.031	59.547	0.410	2021.8	8.4
1000	0.4194	68.034	0.041	0.013	1.358	5.454	0.041	0.025	64.310	0.258	2116.5	5.0
1017	0.4504	72.467	-0.056	0.019	0.833	7.840	-0.056	0.056	66.769	0.365	2163.5	6.9
1065	0.5196	76.140	-0.032	0.013	1.853	4.878	-0.032	0.055	72.410	0.238	2267.0	4.2
1097	0.5841	82.268	-0.004	0.015	1.730	5.236	-0.004	0.015	77.945	0.348	2363.0	5.9
1125	0.6563	89.165	0.022	0.019	1.937	6.287	0.022	0.022	83.546	0.221	2455.3	3.6
1200	0.9905	99.111	0.015	0.001	8.959	0.344	0.015	0.024	98.755	0.120	2684.3	1.7
1340	0.9989	108.288	0.344	0.047	0.225	12.915	0.344	1.300	94.311	3.130	2620.4	45.9
1420	0.9998	171.659	-13.223	0.141	0.024	24.857	-13.110	8.719	127.877	17.051	3055.3	196.3
1600	1.0000	581.145	-74.191	1.995	0.005	102.411	-70.773	56.581	-13.368	73.719	-1123.1	8612.8
Integrated		78.064	-0.005	0.009	26.809	3.553	-0.005	0.019	75.275	0.108	2317.4	7.8

MLM 76 MICROCLINE BT-1F
Mass=0.0044 g

Weighted average of J from standards = 0.027660 +/- 0.000074

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-6 ml/g	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
580	0.0062	461.345	-0.416	0.085	0.103	5.442	-0.416	0.247	436.107	3.651	4636.1	13.9
640	0.0247	53.921	0.091	0.018	0.305	9.888	0.091	0.125	48.578	2.474	1536.6	52.7
690	0.0656	36.773	0.013	0.009	0.676	7.166	0.013	0.026	34.123	0.351	1199.1	9.0
730	0.1054	44.499	-0.011	0.008	0.656	5.614	-0.011	0.040	41.986	0.356	1390.5	8.2
785	0.1744	47.658	0.019	0.005	1.139	3.023	0.019	0.008	46.202	0.243	1485.3	5.3
830	0.2336	49.992	0.052	0.003	0.977	2.054	0.052	0.020	48.951	0.281	1544.5	6.0
875	0.2844	52.743	0.001	0.005	0.840	2.971	0.001	0.013	51.161	0.683	1590.7	14.1
945	0.3270	59.528	0.070	0.005	0.702	2.606	0.070	0.056	57.964	0.523	1726.1	10.0
980	0.3696	64.328	0.024	0.007	0.704	3.322	0.024	0.025	62.178	0.314	1805.1	5.8
1000	0.3990	69.314	-0.055	0.002	0.485	0.899	-0.055	0.071	68.672	1.995	1920.5	34.3
1030	0.4200	76.333	0.103	0.022	0.346	8.460	0.103	0.094	69.865	1.594	1940.9	27.1
1065	0.4697	78.952	0.017	0.013	0.822	4.756	0.017	0.033	75.183	0.371	2029.2	6.0
1097	0.5061	84.296	0.204	0.019	0.601	6.754	0.204	0.212	78.599	1.609	2083.7	25.3
1125	0.5579	89.873	-0.024	0.019	0.855	6.344	-0.024	0.026	84.155	0.320	2169.0	4.8
1600	1.0000	90.752	-0.006	0.000	7.298	0.106	-0.006	0.003	90.639	0.124	2263.7	1.8
Integrated		76.751	0.012	0.006	16.509	2.273	0.012	0.010	74.992	0.126	2026.1	3.8

MLM 69 DYKE KD-4B3
Mass=0.1716 g

Weighted average of J from standards = 0.027660 +/- 0.000074

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-6 ml/g	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
500	0.0012	173.275	3.390	0.269	0.001	45.678	3.397	0.153	94.327	9.329	2315.5	129.0
750	0.0053	62.963	4.509	0.065	0.004	30.199	4.522	0.087	44.068	2.286	1437.9	51.4
780	0.0258	53.293	3.280	0.015	0.019	7.924	3.287	0.011	49.160	0.392	1548.9	8.3
820	0.0627	56.085	2.680	0.010	0.034	4.899	2.684	0.007	53.415	0.203	1636.7	4.1
840	0.1396	56.249	1.804	0.005	0.072	2.337	1.806	0.005	54.984	0.093	1668.1	1.8
910	0.3956	68.764	2.169	0.001	0.238	0.264	2.172	0.005	68.664	0.083	1920.4	1.4
1000	0.8813	70.647	4.855	0.002	0.452	0.413	4.870	0.007	70.562	0.089	1952.7	1.5
1100	0.9268	63.376	13.068	0.008	0.042	2.371	13.180	0.021	62.389	0.186	1809.0	3.4
1200	0.9854	73.648	51.918	0.019	0.055	2.419	53.735	0.116	74.365	0.215	2015.9	3.5
1400	1.0000	81.528	26.226	0.044	0.014	13.489	26.682	0.066	71.743	0.524	1972.6	8.8
Integrated		68.278	7.353	0.005	0.930	1.449	7.388	0.002	67.597	0.045	1901.9	3.2

MLM 72 DYKE KD15-B12 S.A.
Mass=0.2065 g

Weighted average of J from standards = 0.027660 +/- 0.000074

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-6 ml/g	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
500	0.0673	45.907	0.846	0.011	0.096	7.018	0.846	0.003	42.694	0.081	1406.7	1.9
750	0.2943	66.128	1.220	0.009	0.322	3.833	1.221	0.006	63.629	0.123	1831.6	2.2
900	0.5725	63.252	1.499	0.002	0.395	0.532	1.500	0.003	62.962	0.103	1819.4	1.9
975	0.7899	70.097	1.851	0.001	0.308	0.426	1.853	0.006	69.867	0.114	1941.0	1.9
1050	0.8633	68.446	3.606	0.003	0.104	0.721	3.614	0.009	68.096	0.111	1910.6	1.9
1110	0.9188	78.708	23.787	0.011	0.079	1.871	24.161	0.027	78.435	0.082	2081.1	1.3
1300	0.9943	84.004	15.090	0.007	0.107	1.153	15.240	0.038	83.844	0.154	2164.4	2.3
1600	1.0000	86.738	9.356	0.037	0.008	11.664	9.414	0.061	77.077	0.440	2059.6	7.0
Integrated		67.187	3.954	0.005	1.419	1.785	3.964	0.002	66.142	0.040	1876.4	3.2

NHD BIO #1 MICRO-GABBRO DIKE
Single grain

Weighted average of J from standards = 0.024922 +/- 0.000135

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-12 ml	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
300	0.0042	157.809	-0.544	-0.075	0.039	-13.973	-0.544	0.274	179.764	48.313	3069.0	396.4
600	0.1478	136.284	0.011	-0.001	1.330	-0.297	0.011	0.007	136.660	1.550	2675.3	15.8
700	0.3243	157.139	-0.001	-0.001	1.634	-0.232	-0.001	0.005	157.475	1.565	2876.1	14.3
800	0.5644	162.664	0.016	0.004	2.223	0.648	0.016	0.006	161.583	1.305	2913.3	11.7
850	0.6982	158.680	0.013	0.001	1.239	0.237	0.013	0.013	158.277	1.664	2883.4	15.1
900	0.8492	156.196	0.024	-0.001	1.399	-0.183	0.024	0.013	156.455	1.722	2866.8	15.8
950	0.8992	158.288	0.039	0.001	0.463	0.142	0.039	0.030	158.038	2.832	2881.3	25.8
1000	0.9257	160.661	-0.012	-0.003	0.245	-0.487	-0.012	0.075	161.413	8.855	2911.7	79.3
1100	0.9526	164.210	0.145	-0.009	0.249	-1.687	0.145	0.084	166.966	9.384	2960.8	81.7
1200	0.9716	159.319	0.202	0.061	0.176	11.310	0.202	0.104	141.294	11.061	2722.0	110.0
1350	0.9927	156.506	0.012	0.038	0.195	7.103	0.012	0.088	145.364	9.526	2762.0	92.6
1600	1.0000	174.326	0.133	0.056	0.067	9.565	0.133	0.274	157.638	36.773	2877.6	335.5
Integrated		156.032	0.019	0.002	9.259	0.423	0.019	0.006	155.345	0.819	2856.6	10.8

NHD BIO #2 MICRO-GABBRO DIKE
Single grain

Weighted average of J from standards = 0.024922 +/- 0.000135

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-12 ml	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
400	0.0097	88.846	-0.004	0.028	0.126	9.250	-0.004	0.080	80.602	12.789	1987.2	191.1
500	0.0554	53.596	0.046	0.015	0.589	8.499	0.046	0.015	49.016	2.601	1440.0	52.6
550	0.0885	129.954	0.031	0.009	0.427	1.992	0.031	0.021	127.340	3.578	2577.6	38.5
600	0.1451	156.466	0.028	0.006	0.730	1.193	0.028	0.019	154.573	2.940	2849.4	27.2
650	0.3441	153.741	0.004	0.005	2.568	0.869	0.004	0.004	152.376	1.047	2829.0	9.8
700	0.3939	164.384	0.024	0.012	0.642	2.154	0.024	0.019	160.817	3.247	2906.4	29.1
750	0.4723	164.297	0.016	0.003	1.011	0.598	0.016	0.009	163.288	2.188	2928.4	19.4
800	0.5714	164.588	0.007	0.002	1.279	0.316	0.007	0.009	164.040	2.093	2935.1	18.5
850	0.7356	162.738	0.016	0.005	2.119	0.930	0.016	0.008	161.197	1.629	2909.8	14.6
900	0.8302	160.249	0.032	0.005	1.221	0.949	0.032	0.010	158.702	1.963	2887.3	17.8
950	0.8711	156.515	0.074	-0.010	0.527	-1.929	0.074	0.017	159.512	3.832	2894.6	34.6
1100	0.9435	159.228	0.048	0.008	0.935	1.479	0.048	0.012	156.849	2.009	2870.4	18.4
1300	0.9849	157.749	0.109	0.012	0.534	2.265	0.109	0.020	154.158	3.540	2845.6	32.9
1600	1.0000	162.344	-0.003	-0.017	0.195	-3.096	-0.003	0.061	167.340	9.110	2964.0	79.2
Integrated		153.233	0.025	0.005	12.904	1.019	0.025	0.003	151.646	0.631	2822.1	9.7

NHD HBD #1 MICRO-GABBRO DIKE
Single grain

Weighted average of J from standards = 0.024922 +/- 0.000135

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-12 ml	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
Fuse	1.0000	194.497	3.756	0.003	5.381	0.312	3.765	0.011	194.336	0.534	3184.7	4.1
Integrated		194.497	3.756	0.003	5.381	0.312	3.765	0.011	194.336	0.534	3184.7	9.1

NHD HBD #2 MICRO-GABBRO DIKE
Single grain

Weighted average of J from standards = 0.024922 +/- 0.000135

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-12 ml	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
700	0.0012	198.182	13.397	-0.199	0.016	-30.182	13.515	6.194	260.230	165.980	3631.5	997.0
800	0.0047	187.431	3.523	-0.022	0.047	-3.535	3.531	0.526	194.472	44.545	3185.8	342.6
900	0.1095	194.983	3.800	0.004	1.385	0.520	3.809	0.039	194.422	2.248	3185.4	17.3
950	0.7496	193.978	3.850	0.002	8.466	0.191	3.859	0.019	194.065	0.854	3182.6	6.6
1000	0.8904	192.135	3.850	0.005	1.862	0.686	3.860	0.030	191.268	1.611	3161.0	12.6
1050	0.9082	199.219	3.964	0.033	0.235	4.682	3.974	0.104	190.355	8.858	3153.8	69.3
1200	1.0000	196.399	4.026	0.015	1.214	2.042	4.037	0.040	192.866	2.572	3173.4	19.9
Integrated		194.121	3.873	0.004	13.225	0.498	3.883	0.015	193.613	0.736	3179.1	9.9

NHD HBD #3 MICRO-GABBRO DIKE
Single grain

Weighted average of J from standards = 0.024922 +/- 0.000135

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-12 ml	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
800	0.0345	184.073	5.062	-0.007	0.139	-1.297	5.079	0.232	187.048	14.750	3127.7	117.1
900	0.4543	193.033	4.313	0.007	1.684	0.836	4.325	0.032	191.929	1.542	3166.1	12.0
925	0.8582	192.801	4.140	0.005	1.620	0.554	4.151	0.036	192.221	1.843	3168.4	14.3
950	0.8694	134.648	4.256	0.077	0.045	16.687	4.268	0.604	112.467	40.559	2409.8	479.5
1000	0.8797	156.045	4.894	-0.056	0.041	-10.887	4.909	0.921	173.554	50.928	3017.3	430.0
1200	1.0000	189.513	5.811	0.008	0.483	0.945	5.833	0.109	188.406	4.725	3138.5	37.3
Integrated		191.173	4.454	0.006	4.012	0.690	4.467	0.027	190.378	1.448	3154.0	13.9

NHD HBD #4 MICRO-GABBRO DIKE
Single grain

Weighted average of J from standards = 0.024922 +/- 0.000135

Temperature (Deg C)	Cumulative 39Ar	40Ar/39Ar measured	37Ar/39Ar measured	36Ar/39Ar measured	Volume 39Ar x 10-12 ml	% Atmospheric 40Ar	37Ca/39K	+/-	40*/39K	+/-	Age (Ma)	+/- (Ma)
800	0.0473	193.139	3.708	0.037	0.181	5.552	3.717	0.164	182.830	12.930	3093.9	104.6
900	0.6586	193.363	3.822	0.002	2.337	0.174	3.832	0.035	193.480	1.838	3178.1	14.2
1200	1.0000	195.064	3.889	0.007	1.305	0.852	3.899	0.046	193.865	2.246	3181.1	17.3
Integrated		193.933	3.840	0.005	3.823	0.660	3.849	0.028	193.108	1.494	3175.2	14.1